# CCS6224
# Network Security

## Lecture 11
## IP Security

# IP Security Overview

- RFC 1636
  - "Security in the Internet Architecture"
  - Issued in 1994 by the Internet Architecture Board (IAB)
  - Identifies key areas for security mechanisms
    - Need to secure the network infrastructure from unauthorized monitoring and control of network traffic
    - Need to secure end-user-to-end-user traffic using authentication and encryption mechanisms
  - IAB included authentication and encryption as necessary security features in the next generation IP (IPv6)
    - The IPsec specification now exists as a set of Internet standards

# Applications of IPsec

- IPsec provides the capability to secure communications across a LAN, private and public WANs, and the Internet

Examples include:

- Secure branch office connectivity over the Internet
- Secure remote access over the Internet
- Establishing extranet and intranet connectivity with partners
- Enhancing electronic commerce security

- Principal feature of IPsec is that it can encrypt and/or authenticate all traffic at the IP level
  - Thus all distributed applications (remote logon, client/server, e-mail, file transfer, Web access) can be secured

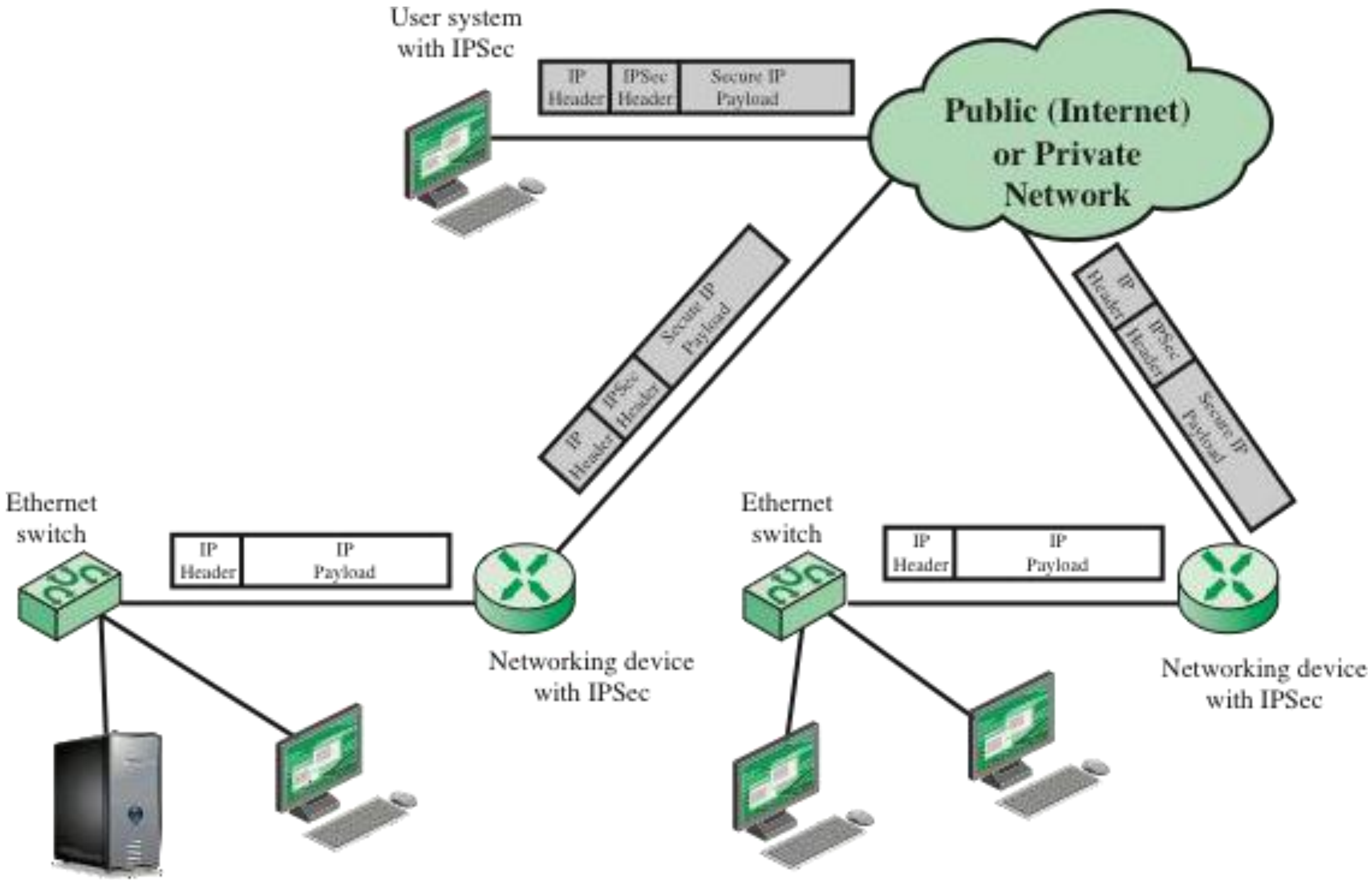



**Figure 20.1 An IP Security Scenario**

# Benefits of IPSec

- Some of the benefits of IPsec:
  - When IPsec is implemented in a firewall or router, it provides strong security that can be applied to all traffic crossing the perimeter
    - Traffic within a company or workgroup does not incur the overhead of security-related processing
  - IPsec in a firewall is resistant to bypass if all traffic from the outside must use IP and the firewall is the only means of entrance from the Internet into the organization
  - IPsec is below the transport layer (TCP, UDP) and so is transparent to applications
    - There is no need to change software on a user or server system when IPsec is implemented in the firewall or router
  - IPsec can be transparent to end users
    - There is no need to train users on security mechanisms, issue keying material on a per-user basis, or revoke keying material when users leave the organization
  - IPsec can provide security for individual users if needed
    - This is useful for offsite workers and for setting up a secure virtual subnetwork within an organization for sensitive applications

# Routing Applications

› IPsec can play a vital role in the routing architecture required for internetworking

## IPsec can assure that:

- A router advertisement comes from an authorized router
- A router seeking to establish or maintain a neighbor relationship with a router in another routing domain is an authorized router
- A redirect message comes from the router to which the initial IP packet was sent
- A routing update is not forged

# IPsec Documents

## Encapsulating Security Payload (ESP)

- Consists of an encapsulating header and trailer used to provide encryption or combined encryption/authentication
- The current specification is RFC 4303, *IP Encapsulating Security Payload (ESP)*

## Internet Key Exchange (IKE)

- A collection of documents describing the key management schemes for use with IPsec
- The main specification is RFC 5996, *Internet Key Exchange (IKEv2) Protocol,* but there are a number of related RFCs

## Authentication Header (AH)

- An extension header to provide message authentication
- The current specification is RFC 4302, *IP Authentication Header*

## Cryptographic algorithms

- This category encompasses a large set of documents that define and describe cryptographic algorithms for encryption, message authentication, pseudorandom functions (PRFs), and cryptographic key exchange

## Architecture

- Covers the general concepts, security requirements, definitions, and mechanisms defining IPsec technology
- The current specification is RFC4301, *Security Architecture for the Internet Protocol*

## Other

- There are a variety of other IPsec-related RFCs, including those dealing with security policy and management information base (MIB) content

# IPsec Services

- IPsec provides security services at the IP layer by enabling a system to:
  - Select required security protocols
  - Determine the algorithm(s) to use for the service(s)
  - Put in place any cryptographic keys required to provide the requested services

- RFC 4301 lists the following services:
  - Access control
  - Connectionless integrity
  - Data origin authentication
  - Rejection of replayed packets (a form of partial sequence integrity)
  - Confidentiality (encryption)
  - Limited traffic flow confidentiality

# Transport and Tunnel Modes

## TRANSPORT MODE

- Provides protection primarily for upper-layer protocols
  - Examples include a TCP or UDP segment or an ICMP packet
- Typically used for end-to-end communication between two hosts
- ESP in transport mode encrypts and optionally authenticates the IP payload but not the IP header
- AH in transport mode authenticates the IP payload and selected portions of the IP header

## TUNNEL MODE

- Provides protection to the entire IP packet
- Used when one or both ends of a security association (SA) are a security gateway
- A number of hosts on networks behind firewalls may engage in secure communications without implementing IPsec
- ESP in tunnel mode encrypts and optionally authenticates the entire inner IP packet, including the inner IP header
- AH in tunnel mode authenticates the entire inner IP packet and selected portions of the outer IP header

# Tunnel Mode and Transport Mode Functionality

| | Transport Mode SA | Tunnel Mode SA |
|---|---|---|
| AH | Authenticates IP payload and selected portions of IP header and IPv6 extension headers. | Authenticates entire inner IP packet (inner header plus IP payload) plus selected portions of outer IP header and outer IPv6 extension headers. |
| ESP | Encrypts IP payload and any IPv6 extension headers following the ESP header. | Encrypts entire inner IP packet. |
| ESP with Authentication | Encrypts IP payload and any IPv6 extension headers following the ESP header. Authenticates IP payload but not IP header. | Encrypts entire inner IP packet. Authenticates inner IP packet. |

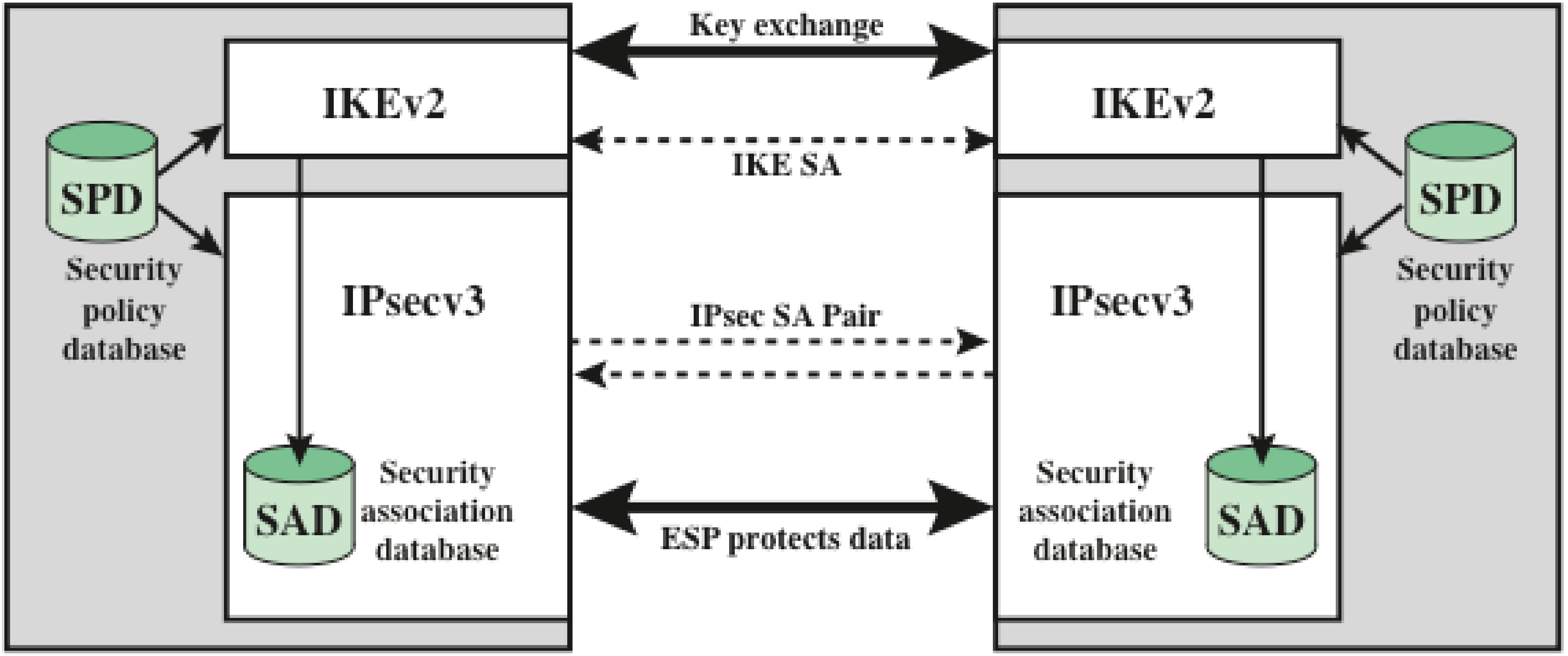



Figure 20.2 IPsec Architecture

# Security Association (SA)

- A one-way logical connection between a sender and a receiver that affords security services to the traffic carried on it
- In any IP packet, the SA is uniquely identified by the Destination Address in the IPv4 or IPv6 header and the SPI in the enclosed extension header (AH or ESP)

Uniquely identified by three parameters:

# Security Association Database (SAD)

- Defines the parameters associated with each SA
- Normally defined by the following parameters in a SAD entry:
  - Security parameter index
  - Sequence number counter
  - Sequence counter overflow
  - Anti-replay window
  - AH information
  - ESP information
  - Lifetime of this security association
  - IPsec protocol mode
  - Path MTU

**Textbook page 669**

# Security Policy Database (SPD)

- The means by which IP traffic is related to specific SAs
  - Contains entries, each of which defines a subset of IP traffic and points to an SA for that traffic
- In more complex environments, there may be multiple entries that potentially relate to a single SA or multiple SAs associated with a single SPD entry
  - Each SPD entry is defined by a set of IP and upper-layer protocol field values called *selectors*
  - These are used to filter outgoing traffic in order to map it into a particular SA

# SPD Entries

- The following selectors determine an SPD entry:

**Remote IP address**

- This may be a single IP address, an enumerated list or range of addresses, or a wildcard (mask) address
- The latter two are required to support more than one destination system sharing the same SA

**Local IP address**

- This may be a single IP address, an enumerated list or range of addresses, or a wildcard (mask) address
- The latter two are required to support more than one source system sharing the same SA

**Next layer protocol**

- The IP protocol header includes a field that designates the protocol operating over IP

**Name**

- A user identifier from the operating system
- Not a field in the IP or upper-layer headers but is available if IPsec is running on the same operating system as the user

**Local and remote ports**

- These may be individual TCP or UDP port values, an enumerated list of ports, or a wildcard port

# Host SPD Example

| Protocol | Local IP | Port | Remote IP | Port | Action | Comment |
|---|---|---|---|---|---|---|
| UDP | 1.2.3.101 | 500 | * | 500 | BYPASS | IKE |
| ICMP | 1.2.3.101 | * | * | * | BYPASS | Error messages |
| * | 1.2.3.101 | * | 1.2.3.0/24 | * | PROTECT: ESP intransport-mode | Encrypt intranet traffic |
| TCP | 1.2.3.101 | * | 1.2.4.10 | 80 | PROTECT: ESP intransport-mode | Encrypt to server |
| TCP | 1.2.3.101 | * | 1.2.4.10 | 443 | BYPASS | TLS: avoid double encryption |
| * | 1.2.3.101 | * | 1.2.4.0/24 | * | DISCARD | Others in DMZ |
| * | 1.2.3.101 | * | * | * | BYPASS | Internet |

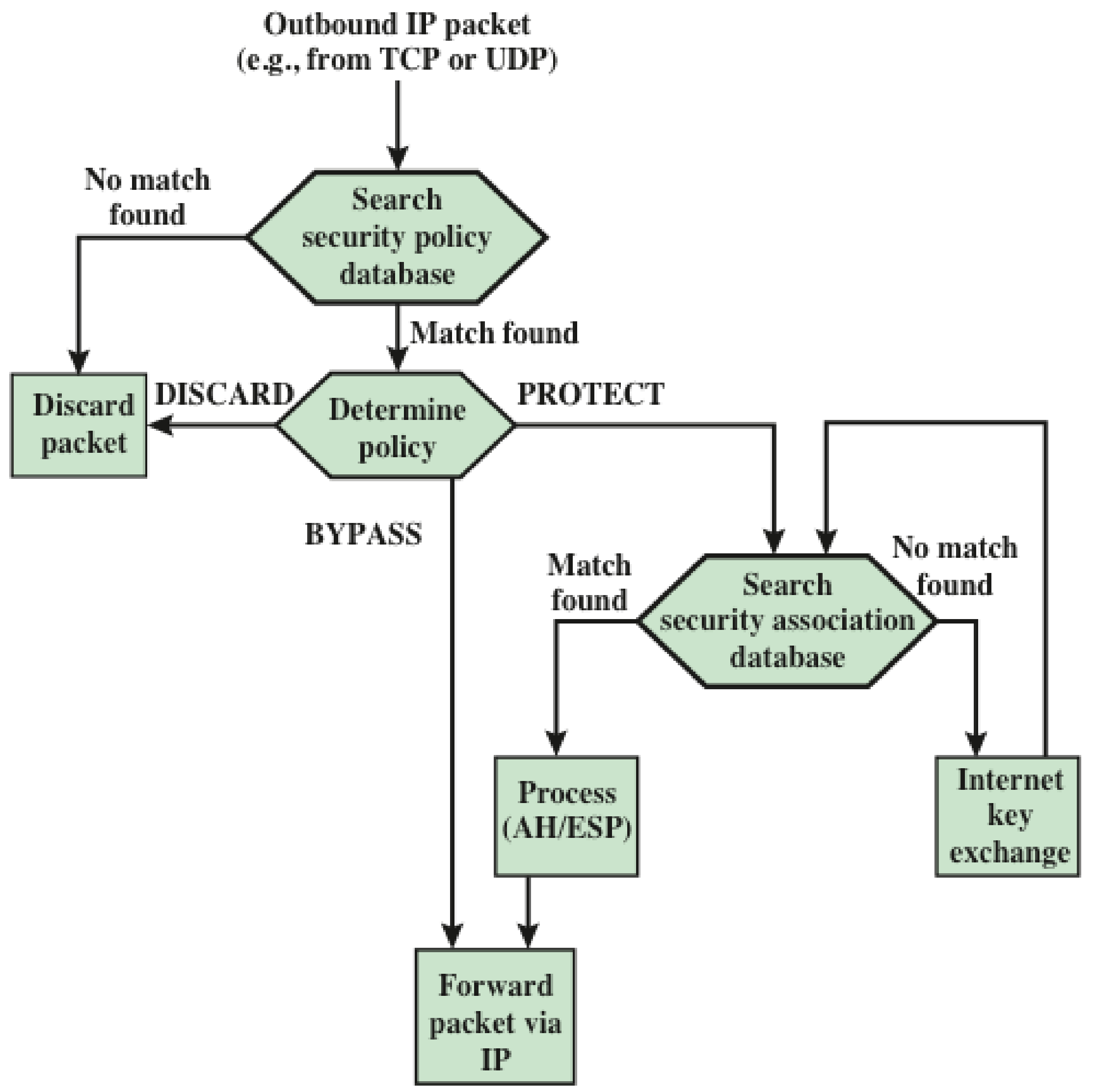



**Figure 20.3 Processing Model for Outbound Packets**

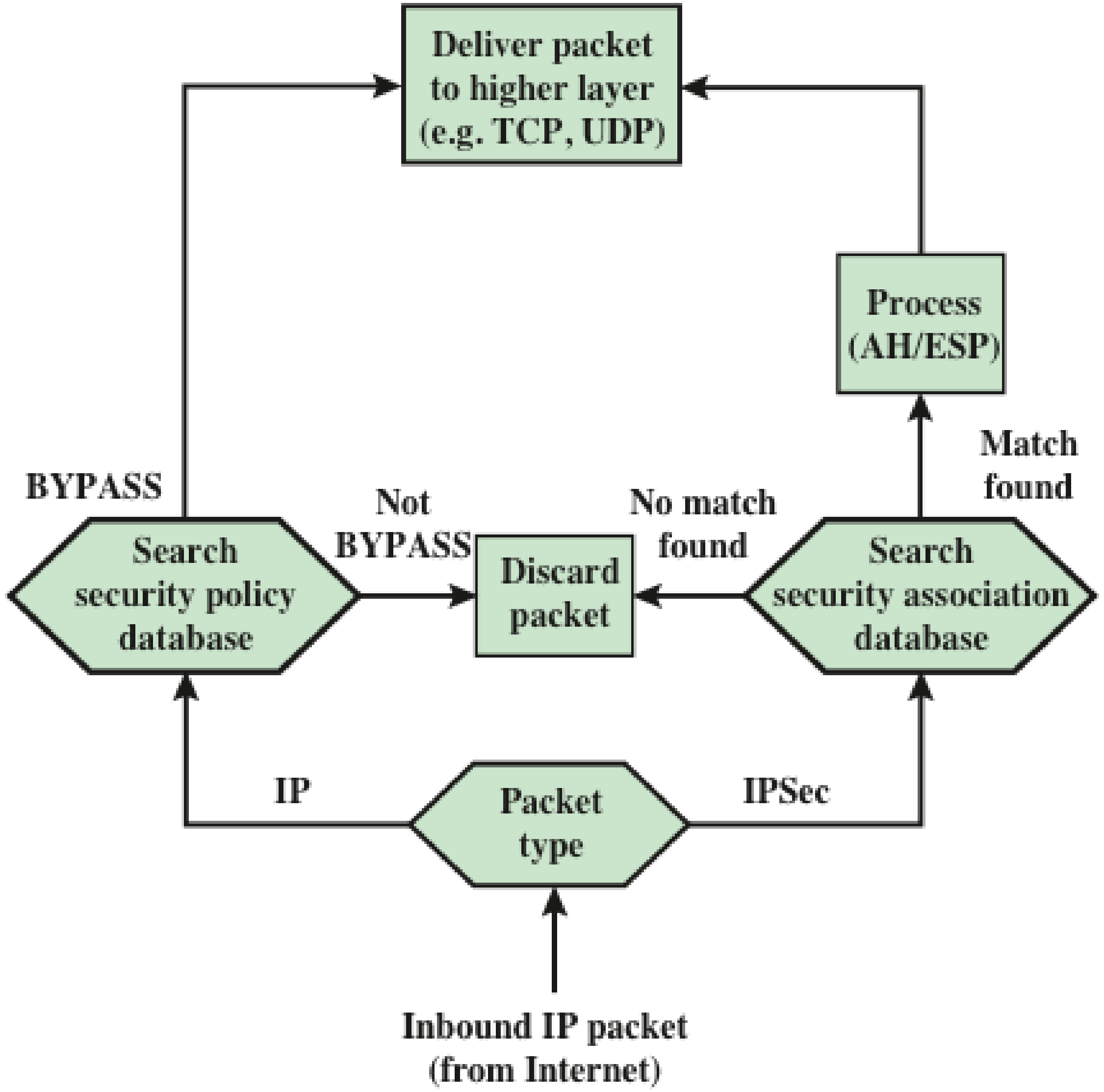



**Figure 20.4 Processing Model for Inbound Packets**

# Encapsulating Security Payload (ESP)

- Used to encrypt the Payload Data, Padding, Pad Length, and Next Header fields
  - If the algorithm requires cryptographic synchronization data then these data may be carried explicitly at the beginning of the Payload Data field
- An optional ICV field is present only if the integrity service is selected and is provided by either a separate integrity algorithm or a combined mode algorithm that uses an ICV
  - ICV is computed after the encryption is performed
  - This order of processing facilitates reducing the impact of DoS attacks
  - Because the ICV is not protected by encryption, a keyed integrity algorithm must be employed to compute the ICV
- The Padding field serves several purposes:
  - If an encryption algorithm requires the plaintext to be a multiple of some number of bytes, the Padding field is used to expand the plaintext to the required length
  - Used to assure alignment of Pad Length and Next Header fields
  - Additional padding may be added to provide partial traffic-flow confidentiality by concealing the actual length of the payload

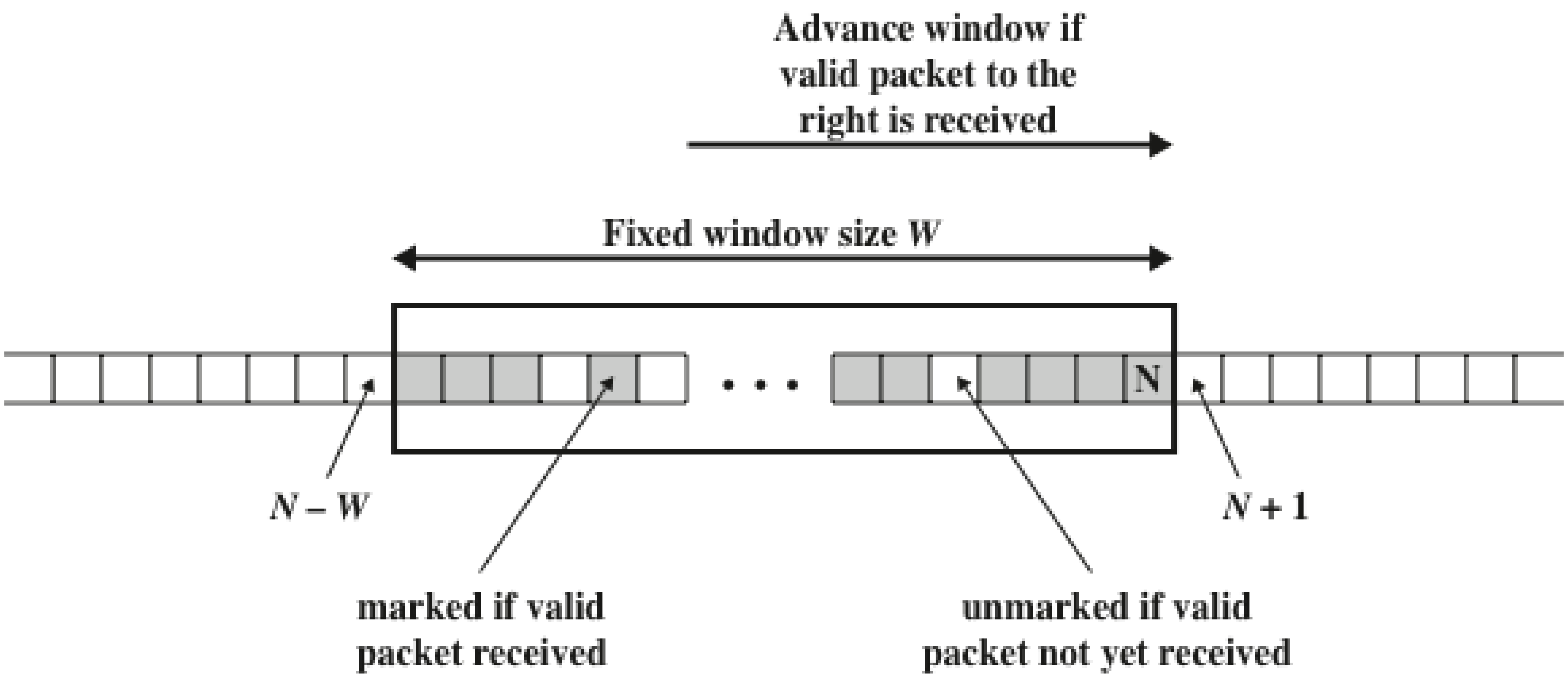
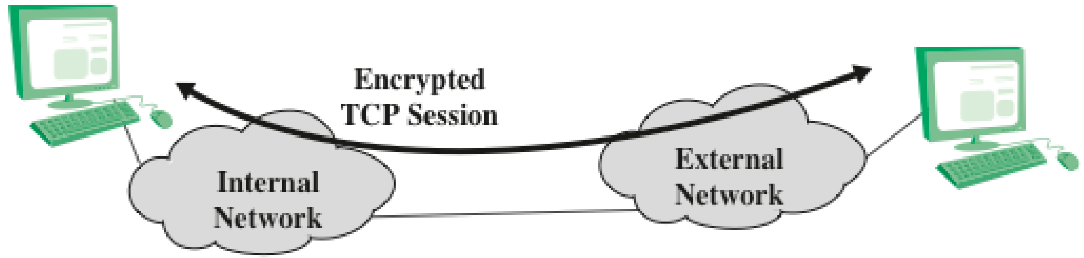
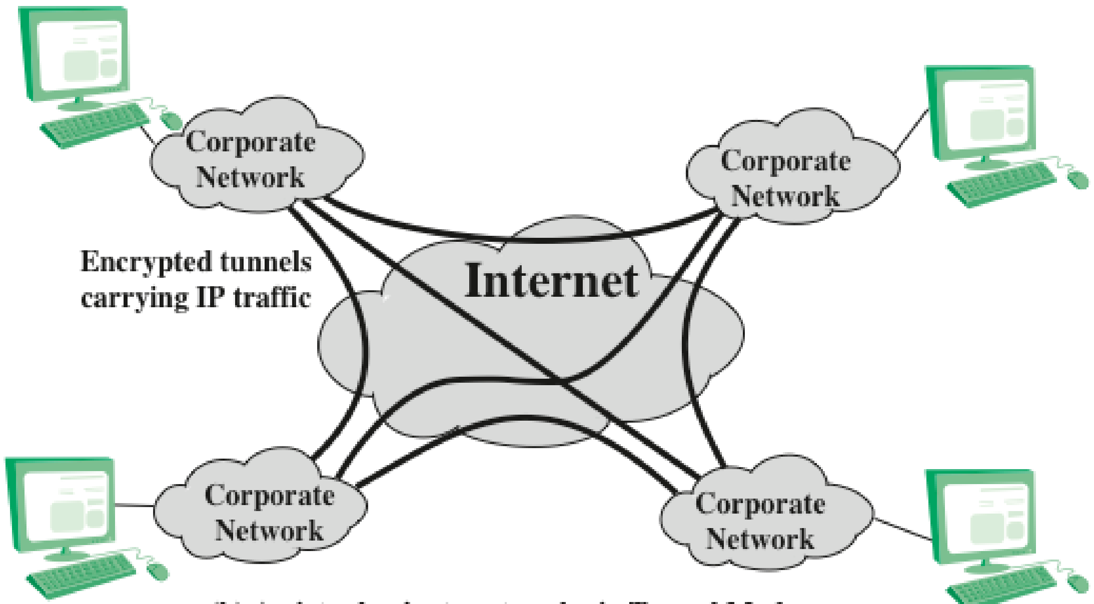



Figure 20.6 Anti-Replay Mechanism




(a) Transport-level security




(b) A virtual private network via Tunnel Mode

**Figure 20.7 Transport-Mode vs. Tunnel-Mode Encryption**

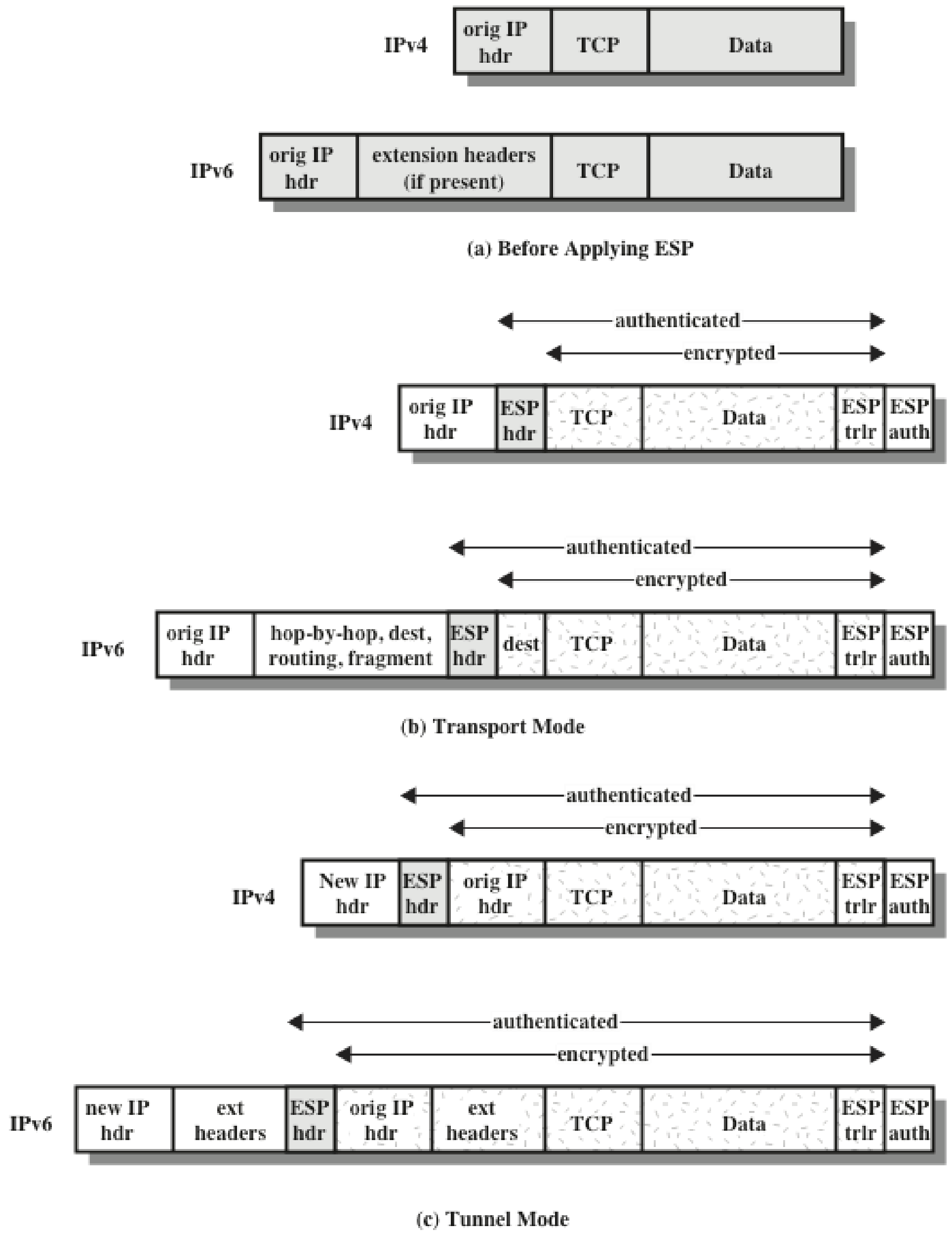
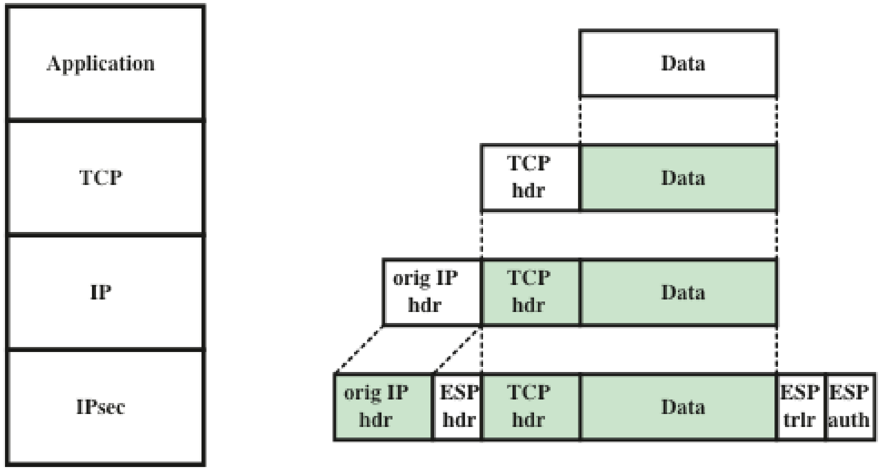
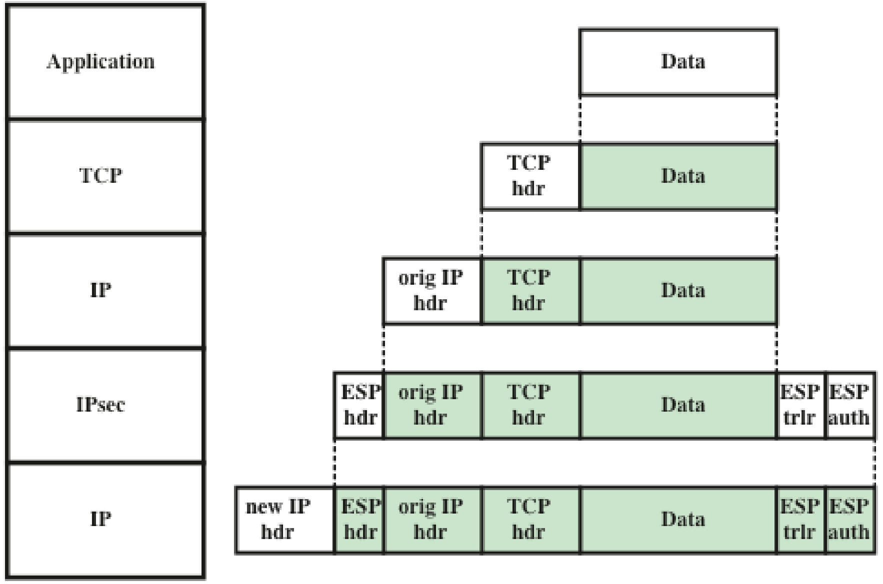



**Figure 20.8 Scope of ESP Encryption and Authentication**




(a) Transport mode




(b) Tunnel mode

Figure 20.9 Protocol Operation for ESP

# Combining Security Associations

- An individual SA can implement either the AH or ESP protocol but not both
- *Security association bundle*
  - Refers to a sequence of SAs through which traffic must be processed to provide a desired set of IPsec services
  - The SAs in a bundle may terminate at different endpoints or at the same endpoint
- May be combined into bundles in two ways:

**Transport adjacency**

- Refers to applying more than one security protocol to the same IP packet without invoking tunneling
- This approach allows for only one level of combination

**Iterated tunneling**

- Refers to the application of multiple layers of security protocols effected through IP tunneling
- This approach allows for multiple levels of nesting

# ESP with Authentication Option

- In this approach, the first user applies ESP to the data to be protected and then appends the authentication data field

**Transport mode ESP**

- Authentication and encryption apply to the IP payload delivered to the host, but the IP header is not protected

**Tunnel mode ESP**

- Authentication applies to the entire IP packet delivered to the outer IP destination address and authentication is performed at that destination
- The entire inner IP packet is protected by the privacy mechanism for delivery to the inner IP destination

– For both cases authentication applies to the ciphertext rather than the plaintext

**https://www.youtube.com/watch?v=j-UCxj_1A-c&t=851s**

# Transport Adjacency

- Another way to apply authentication after encryption is to use two bundled transport SAs, with the inner being an ESP SA and the outer being an AH SA
  - In this case ESP is used without its authentication option
  - Encryption is applied to the IP payload
  - AH is then applied in transport mode
  - Advantage of this approach is that the authentication covers more fields
  - Disadvantage is the overhead of two SAs versus one SA

# Transport-Tunnel Bundle

- The use of authentication prior to encryption might be preferable for several reasons:
  - It is impossible for anyone to intercept the message and alter the authentication data without detection
  - It may be desirable to store the authentication information with the message at the destination for later reference
- One approach is to use a bundle consisting of an inner AH transport SA and an outer ESP tunnel SA
  - Authentication is applied to the IP payload plus the IP header
  - The resulting IP packet is then processed in tunnel mode by ESP
    - The result is that the entire authenticated inner packet is encrypted and a new outer IP header is added

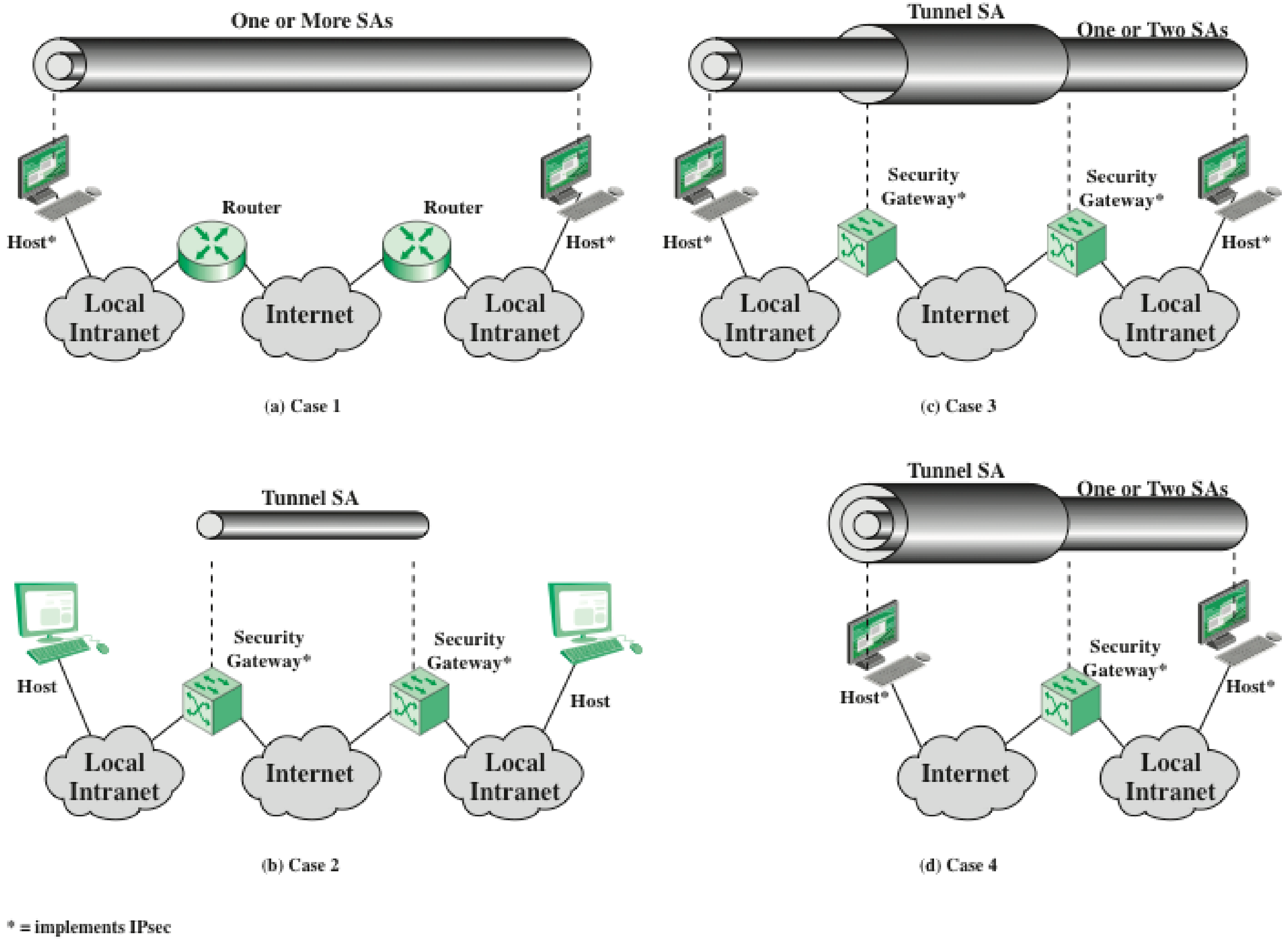
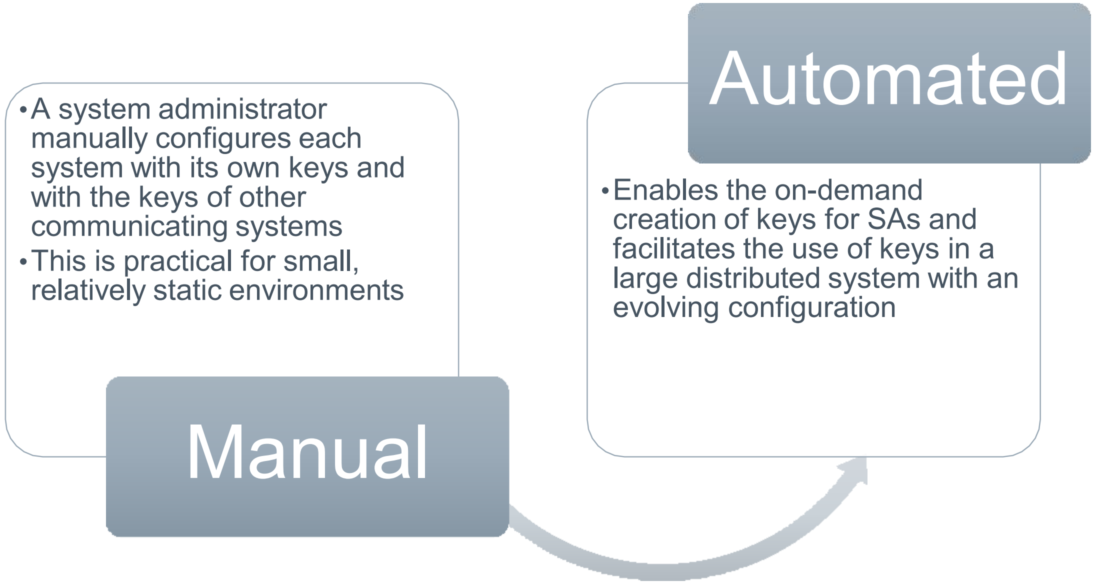



**Figure 20.10 Basic Combinations of Security Associations**

# Internet Key Exchange

- The key management portion of IPsec involves the determination and distribution of secret keys
  - A typical requirement is four keys for communication between two applications
    - Transmit and receive pairs for both integrity and confidentiality
- The IPsec Architecture document mandates support for two types of key management:



# ISAKMP/Oakley

- The default automated key management protocol of IPsec
- Consists of:
  - Oakley Key Determination Protocol
    - A key exchange protocol based on the Diffie-Hellman algorithm but providing added security
    - Generic in that it does not dictate specific formats
  - Internet Security Association and Key Management Protocol (ISAKMP)
    - Provides a framework for Internet key management and provides the specific protocol support, including formats, for negotiation of security attributes
    - Consists of a set of message types that enable the use of a variety of key exchange algorithms

# Features of IKE Key Determination

› Algorithm is characterized by five important features:

1. It employs a mechanism known as cookies to thwart clogging attacks
2. It enables the two parties to negotiate a group; this, in essence, specifies the global parameters of the Diffie-Hellman key exchange
3. It uses nonces to ensure against replay attacks
4. It enables the exchange of Diffie-Hellman public key values
5. It authenticates the Diffie-Hellman exchange to thwart man-in-the-middle-attacks

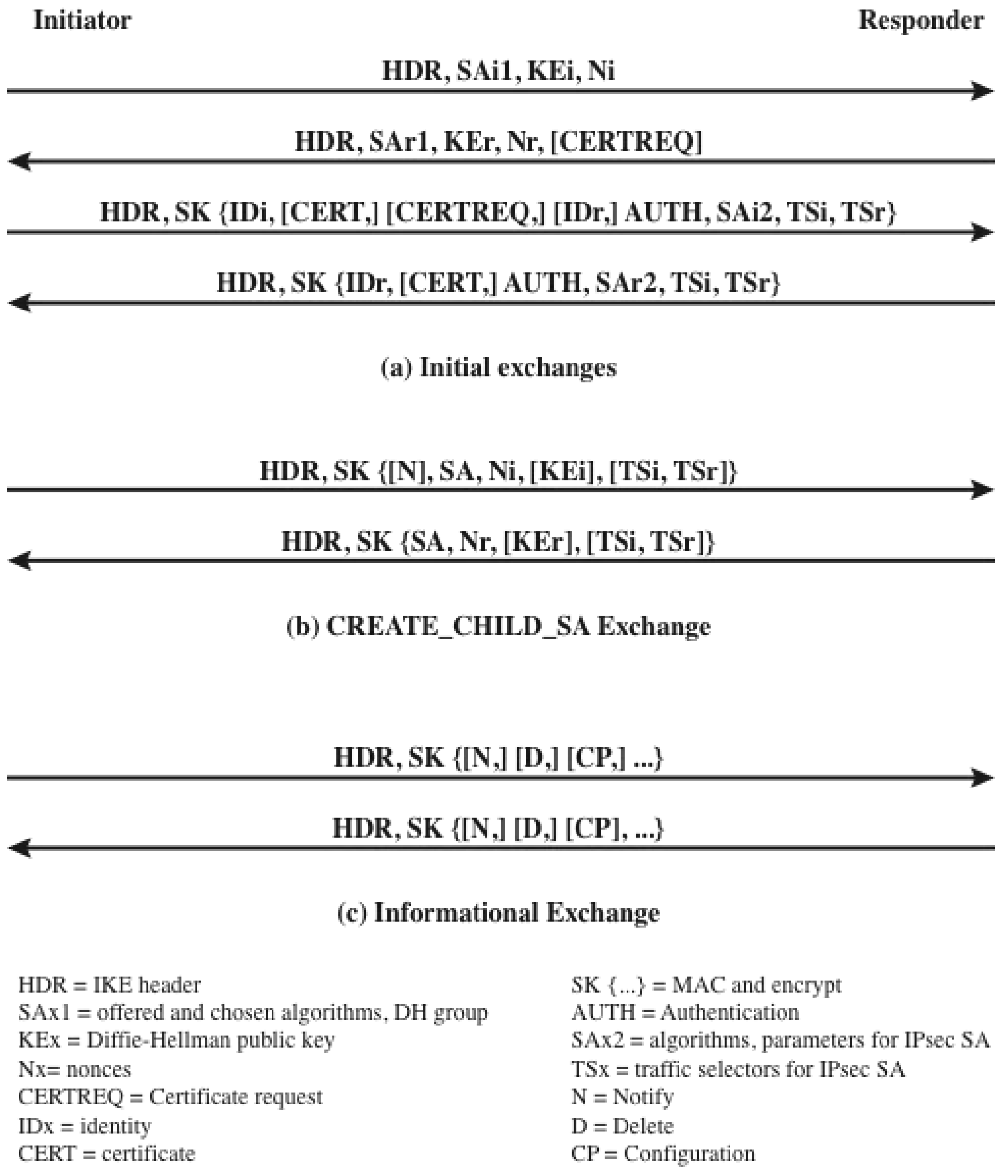



**Figure 20.11 IKEv2 Exchanges**

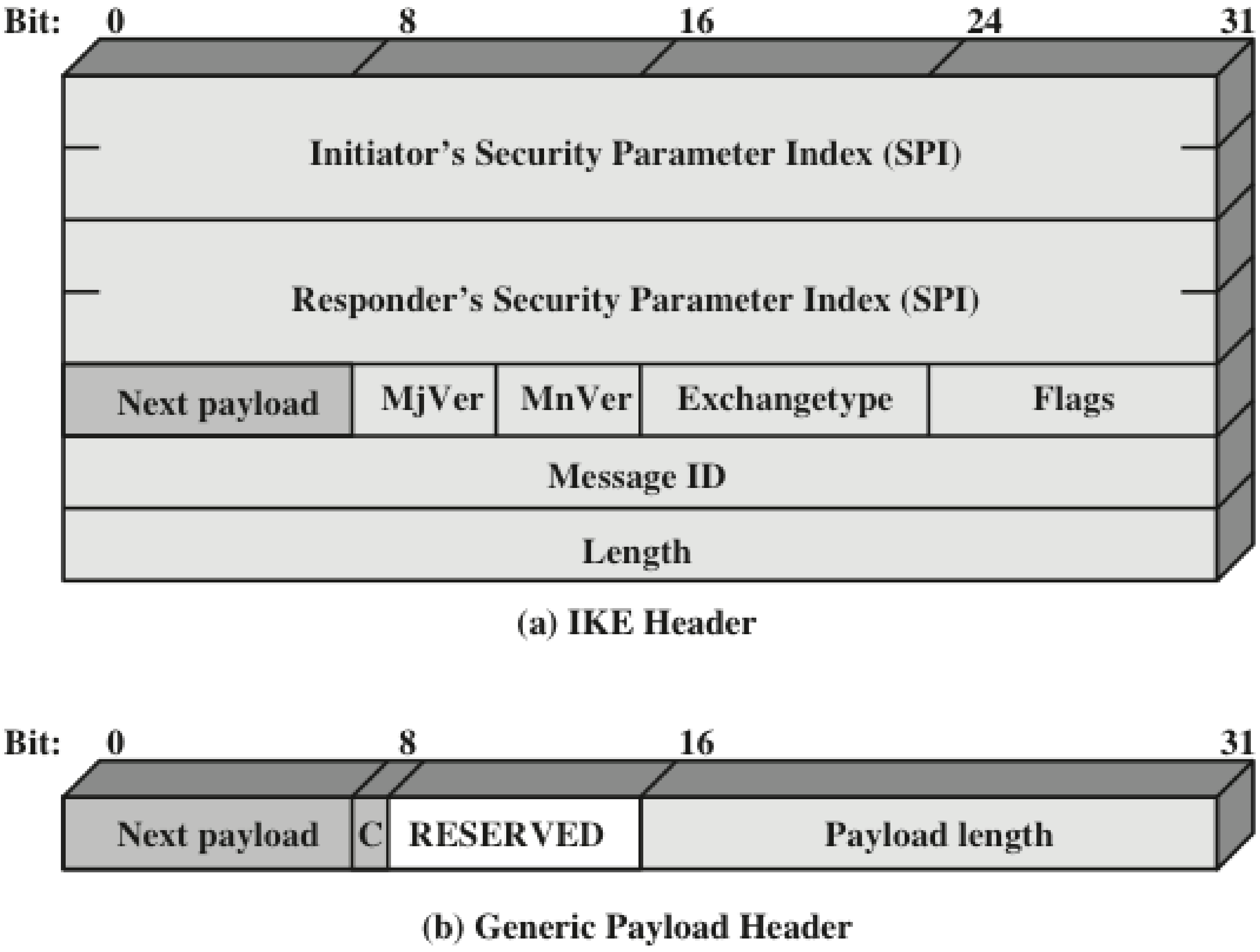



Figure 20.12 IKE Formats

# IKE Payload Types

| Type | Parameters |
|---|---|
| Security Association | Proposals |
| Key Exchange | DH Group #, Key Exchange Data |
| Identification | ID Type, ID Data |
| Certificate | Cert Encoding, Certificate Data |
| Certificate Request | Cert Encoding, Certification Authority |
| Authentication | Auth Method, Authentication Data |
| Nonce | Nonce Data |
| Notify | Protocol-ID, SPI Size, Notify Message Type, SPI, Notification Data |
| Delete | Protocol-ID, SPI Size, # of SPIs, SPI (one or more) |
| Vendor ID | Vendor ID |
| Traffic Selector | Number of TSs, Traffic Selectors |
| Encrypted | IV, Encrypted IKE payloads, Padding, Pad Length, ICV |
| Configuration | CFG Type, Configuration Attributes |
| Extensible Authentication Protocol | EAP Message |